

EXPLORATORY DATA ANALYSIS

HealthConnect Appointment Attendance and No-Show Analysis

Dataset: HealthConnect Appointment Data

By

Ibtihaj Shuaib

Programme: Data Analytics Internship, AnalystLab Africa

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PART 1: WEEK 4 REVIEW

Problem Defined

HealthConnect Clinic is experiencing missed patient appointments, which can result in underutilized appointment slots and challenges in managing clinic resources effectively. The clinic also needs a clearer understanding of the factors and patterns associated with appointment no-shows.

The broader project question is: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

As a Data Analytics Intern, my specific focus will be to investigate the appointment data to understand patterns and factors associated with patient attendance and no-shows.

Proposed Approach

In week 4, the proposed approach was to:

- Analyze Appointment Outcomes
- Examine Overall Attendance and No-Show Patterns
- Investigate Factors Associated with No-Shows
- Calculate and Evaluate the Proposed KPIs
- Compare Relevant Groups and Identify Patterns
- Develop Business Insights and Recommendations

Relevant HealthConnect Resources

The main HealthConnect resources relevant to use in this analysis include:

1. HealthConnect Appointment Dataset
2. HealthConnect Data Dictionary
3. Power Query for data preparation and quality checks and
4. Power BI for exploratory analysis, KPI development, and dashboard creation.

Key Assumptions and Limitations

Assumptions:

- Appointment_outcome accurately represents the recorded outcome of each appointment and can be used as the primary outcome variable.
- The available variables contain sufficient information to investigate patterns associated with appointment attendance and no-shows.

Limitations:

- Some variables contain missing values, including limited missing information in distance_to_clinic_km and waiting_time_minutes.
- Some data types did not initially match the HealthConnect Data Dictionary, requiring data-type corrections during data preparation.

Proposed Focus for Week 5

In Week 5, the focus will be to move from the initial planning and assessment completed in Week 4 into practical analysis. This will involve:

- Preparing and validating the appointment data
- Examining attendance and no-show patterns
- Investigating relationships between relevant variables and appointment outcomes

- Calculating and evaluating selected KPIs, and developing analytical visualization
- Identifying meaningful business insights and recommendations for HealthConnect

PART 2: WEEK 5 PRACTICAL WORK

DATA PREPARATION AND DATA QUALITY ASSESSMENT

Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 appointment records across 18 variables. The variables cover patient characteristics, appointment information, booking information, previous appointment history, reminder activity, operational factors, and appointment outcomes.

The main outcome variable is appointment_outcome, with three categories: Attended, No-Show and Cancelled

Data Quality Checks

The dataset was assessed for: missing values, duplicate records, duplicate appointment IDs, incorrect data types, invalid numerical values, and inconsistent categorical values.

Missing Values

Two variables contained missing observations:

Variable	Missing Records	Approx. Percentage
Distance to clinic	90	1.8%
Waiting time	60	1.2%

The missing values were relatively limited but were considered when interpreting analyses involving distance and waiting time. Reminder_channel did not contain missing values. The

value None was treated as a valid category representing appointments for which no reminder channel was used.

Duplicate Assessment

No completely duplicated records were identified. There were also 5,000 unique appointment IDs across 5,000 records, indicating that there were no duplicate appointment IDs.

Data Type Check

The booking_date and appointment_date fields were initially stored as text. They were converted to proper date fields during data preparation so that they could be used correctly for analysis.

The booking lead days, age group, waiting time and appointment day ordering was also corrected to follow the natural sequence

Validity and Consistency Checks

The numerical variables were checked against the expected ranges:

- Age: 18–80 years
- Booking lead days: 0–60 days
- Previous appointments: 0–11
- Previous no-shows: 0–5
- Distance to clinic: 0.5–45 km among non-missing records
- Waiting time: 2–68 minutes among non-missing records

No obvious invalid or implausible values were identified.

Categorical variables including gender, age group, appointment type, appointment day, reminder status, reminder channel, and appointment outcome were also reviewed and no major unexpected labels were identified.

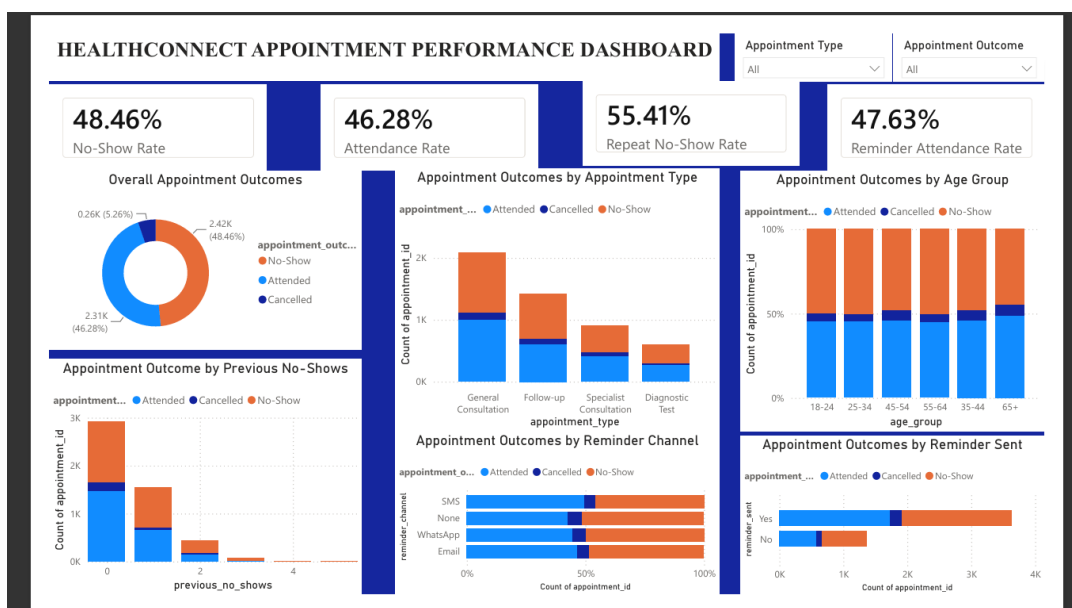
EXPLORATORY DATA ANALYSIS

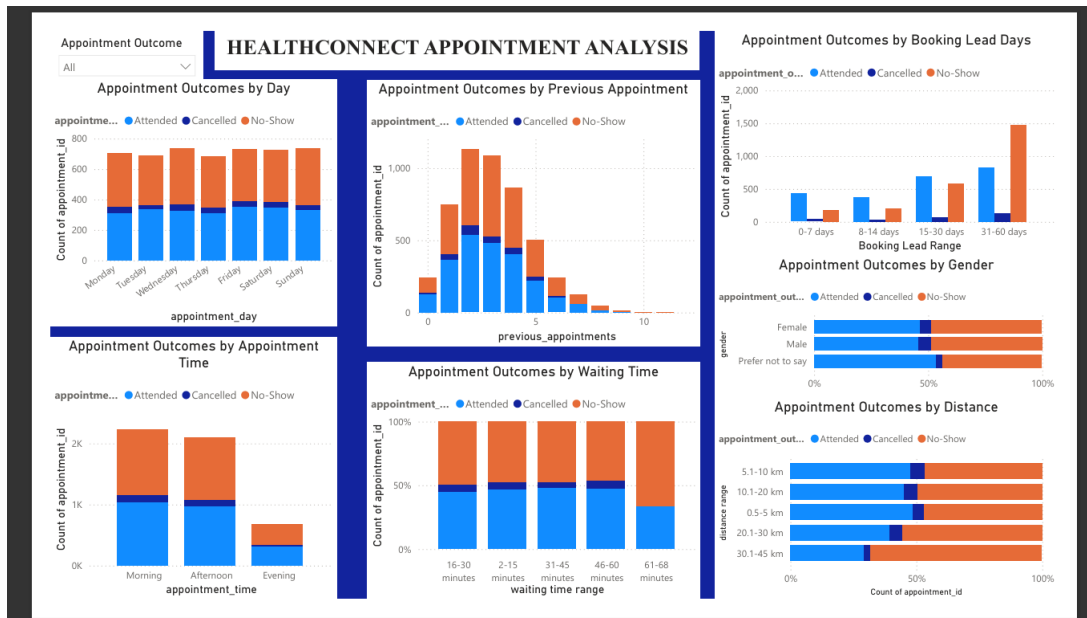
Exploratory Data Analysis Approach

Exploratory analysis was conducted to investigate appointment attendance and no-show patterns across appointment characteristics, patient characteristics, previous appointment history, reminder variables, waiting time, distance to the clinic, and appointment outcomes. The analysis used Power BI visualisations to compare Attended, No-Show, and Cancelled appointments across relevant variables.

Dashboard

An analytical dashboard was developed in Power BI to provide an overview of appointment attendance and no-show patterns. The dashboard contains four KPI cards, an overall appointment outcome visual, and analytical visualisations examining appointment type, previous no-shows, age group, appointment day and time, reminders, booking lead time, waiting time, distance to the clinic, and gender. Two slicers were also included to allow users to filter the analysis by appointment outcome and appointment type.





Analytical Findings

1. Overall Appointment Outcomes

The overall outcome dashboard shows:

- No-Show: 48.46%
- Attended: 46.28%
- Cancelled: 5.26%

The no-show rate is therefore slightly higher than the attendance rate. Almost half of the appointments in the dataset resulted in a no-show, highlighting missed appointments as a major operational concern for HealthConnect.

Interpretation: The relatively high no-show proportion suggests that reducing missed appointments could potentially improve appointment-slot utilisation and overall clinic efficiency.

2. Appointment Outcomes by Appointment Type

General consultations account for the largest number of appointments, followed by follow-up appointments, specialist consultations, and diagnostic tests.

No-shows form a substantial proportion of appointments across all four appointment types. The follow-up category shows a particularly prominent no-show segment relative to its attendance segment.

Interpretation: No-show management should not be restricted to one appointment type. However, follow-up appointments may deserve particular attention because of the relatively large proportion of missed appointments observed in the visualisation.

3. Appointment Outcomes by Appointment Day

Appointment volumes are relatively evenly distributed across the seven days of the week.

Wednesday and Sunday have among the higher total appointment volumes, while Tuesday and Thursday have somewhat lower volumes. No-shows remain a substantial component across all days.

Interpretation: There is no single day that completely explains the no-show problem. This suggests that interventions should not focus exclusively on one weekday.

4. appointment Outcomes by Appointment Time

Morning appointments represent the largest appointment volume, followed closely by afternoon appointments. Evening appointments have considerably fewer appointments. The no-show segment is substantial for both morning and afternoon appointments.

Interpretation: The larger number of morning and afternoon appointments means that these periods account for a significant share of missed appointments. However, the lower volume of evening appointments should be considered when comparing absolute counts.

5. Appointment Outcomes by Age Group

The outcome distribution is generally similar across the age groups. Most age groups have attendance levels around the middle of the distribution, with no-show proportions remaining substantial. The 65+ age group shows the highest attendance proportion and a comparatively lower no-show proportion.

Interpretation: Age does not appear to be as strong a differentiator as previous no-show history, booking lead time, waiting time, or distance. Nevertheless, age-specific engagement strategies may still be useful, particularly for groups showing comparatively lower attendance.

6. Appointment Outcomes by Previous Appointments

The largest volumes of appointments occur among patients with several previous appointments, particularly around 2–4 previous appointments. The no-show component remains present across the different levels of previous appointment history.

Interpretation: The number of previous appointments alone does not appear to provide as clear a distinction as the number of previous no-shows. This suggests that attendance history may be more informative when it specifically captures previous missed appointments.

7. Appointment Outcomes by Previous No-Shows

A strong pattern is visible between previous no-show history and current appointment outcomes. Patients with no previous no-shows form the largest group and have a substantial attendance

component. Among patients with previous no-shows, the no-show component becomes increasingly prominent.

The calculated Repeat No-Show Rate is 55.41% among patients with at least one previous no-show.

Interpretation: Previous no-show history is one of the clearest indicators identified in the analysis. Patients who have previously missed appointments appear more likely to miss another appointment. This makes previous no-show history particularly relevant for targeted appointment management.

8. Appointment Outcomes by Booking Lead Time

Appointments were grouped into:

- 0–7 days;
- 8–14 days;
- 15–30 days; and
- 31–60 days.

The 31–60 day group has the highest appointment volume and the largest number of no-shows. The no-show component also becomes more prominent as booking lead time increases.

Interpretation: Longer booking lead times appear to be associated with greater no-show activity in the dataset. A longer gap between booking and appointment may provide more opportunity for plans to change or for patients to forget the appointment. This makes booking lead time a potentially useful variable for targeted reminders and future predictive analysis.

9. Appointment Outcomes by Reminder Sent

Appointments for which reminders were sent make up the larger group. The calculated Reminder Attendance Rate is 47.63%.

The attendance rate among appointments receiving reminders is only slightly above the overall attendance rate of 46.28%, while the no-show segment remains substantial.

Interpretation: The data does not provide sufficient evidence to conclude that reminders alone substantially reduce no-shows. However, reminders may still have value as part of a broader engagement strategy. The effectiveness of reminders may depend on timing, channel, patient characteristics, or how reminders are targeted.

10. Appointment Outcomes by Reminder Channel

Among the reminder categories, SMS shows the strongest attendance proportion, while WhatsApp shows a somewhat lower attendance proportion. The "None" category has a comparatively lower attendance proportion.

No-show remains substantial across the reminder channels.

Interpretation: SMS appears promising as a reminder channel based on the observed attendance distribution. However, the differences are not sufficient to establish that SMS is causally more effective.

Further analysis could examine reminder timing, frequency, patient characteristics, and no-show history by channel.

11. Appointment Outcomes by Waiting Time

The waiting-time analysis shows relatively similar outcome distributions across the shorter and middle waiting-time groups. However, the longest waiting-time group shows a noticeable decline in attendance and a corresponding increase in the no-show proportion.

Interpretation: Longer waiting times may be associated with poorer appointment outcomes. This suggests that reducing excessive waiting time could potentially improve the patient experience and may also support better appointment attendance.

Further analysis with more detailed operational data would be required before concluding that waiting time directly causes no-shows.

12. Appointment Outcomes by Distance to Clinic

The distance analysis shows a clear pattern in which attendance tends to decrease as distance from the clinic increases. The longest-distance group, particularly the 30–45 km range, has the lowest attendance proportion and a comparatively high no-show proportion.

Interpretation: Distance to the clinic appears to be an important potential barrier to attendance. Patients traveling longer distances may face greater transportation difficulties, cost, time requirements, or other accessibility challenges.

Distance should therefore be considered when designing targeted appointment reminders or patient-support interventions.

13. Appointment Outcomes by Gender

Female and male patients show broadly similar distributions of attendance, cancellations, and no-shows. The Prefer not to say group shows a somewhat higher attendance proportion, although it represents a smaller category.

Interpretation: Gender does not appear to be a major differentiating factor in the overall no-show pattern. Therefore, gender alone may not be an appropriate basis for prioritizing no-show interventions.

KPI DEVELOPMENT

Four KPIs proposed during Week 4 were implemented during Week 5.

No-Show Rate

Definition: This KPI measures the percentage of scheduled appointments that result in a no-show.

Calculation: $\text{No-Show Rate} = \text{No-Show Appointments} \div \text{Total Appointments} \times 100$

Result: 48.46%

Why It Matters: It provides a direct measure of the scale of the missed-appointment problem.

Business Question: How significant is the missed-appointment problem at HealthConnect?

Interpretation: Almost half of all appointments resulted in no-shows, indicating a significant opportunity for improving appointment attendance and clinic resource utilisation.

Attendance Rate

Definition: This KPI measures the percentage of scheduled appointments that are attended.

Calculation: $\text{Attendance Rate} = \text{Attended Appointments} \div \text{Total Appointments} \times 100$

Result: 46.28%

Why It Matters: It provides an overall measure of appointment attendance and complements the No-Show Rate.

Business Question: What proportions of scheduled appointments are successfully attended?

Interpretation: Less than half of all appointments were attended, indicating that attendance improvement should remain an important operational priority.

Repeat No-Show Rate

Definition: This KPI measures the proportion of patients with a history of previous no-shows who subsequently miss another appointment.

Calculation: Repeat No-Show Rate = No-Shows among patients with previous no-shows ÷ Appointments among patients with previous no-shows × 100

Result: 55.41%

Why It Matters: It will help assess whether previous no-show behavior is associated with future appointment attendance.

Business Question: Are patients with previous no-shows more likely to miss another appointment?

Interpretation: The repeat no-show rate of 55.41% indicates a substantial recurrence of missed appointments among patients with previous no-show history. This makes previous no-show history an important variable for targeted intervention.

Reminder Attendance Rate

Definition: This KPI measures the percentage of appointments attended among patients who received an appointment reminder

Calculation: Reminder Attendance Rate = Attended appointments with reminders ÷ Total appointments with reminders × 100

Result: 47.63%

Why It Matters: This provides a basic measure of attendance among appointments receiving reminders.

Business Question: What is the attendance level among appointments where reminders were sent?

Interpretation: The reminder attendance rate of 47.63% is slightly higher than the overall attendance rate of 46.28%. However, the difference is small, and the analysis does not establish a causal effect of reminders.

KEY BUSINESS INSIGHTS

Based on the completed exploratory analysis and KPI results, the following key insights were identified.

1. No-shows represent the largest appointment outcome

The overall no-show rate is 48.46%, compared with an attendance rate of 46.28% and cancellation rate of 5.26%.

Business implication: Missed appointments represent a major operational challenge and a significant opportunity for HealthConnect to improve appointment-slot utilisation.

2. Previous no-show history is a strong risk indicator

Patients with at least one previous no-show have a 55.41% repeat no-show rate.

Business implication: Previous no-show history can be used to identify patients who may benefit from more targeted reminders, confirmation requests, or other engagement strategies.

3. Longer booking lead times are associated with more no-show activity

The 31–60 day booking lead group has the largest appointment volume and the largest number of no-shows, with the no-show segment becoming more prominent at longer lead times.

Business implication: Appointments booked far in advance may require additional reminders or confirmation closer to the appointment date.

4. Longer travel distance is associated with poorer attendance

Attendance decreases across the distance categories, with the longest-distance patients showing the lowest attendance and a higher no-show proportion.

Business implication: Distance may represent an accessibility barrier. HealthConnect could consider targeted communication or alternative support for patients travelling longer distances.

5. Long waiting times are associated with lower attendance

The longest waiting-time group shows noticeably lower attendance and higher no-show representation than the shorter waiting-time groups.

Business implication: Reducing excessive waiting times may improve the patient experience and could potentially support better attendance.

6. Reminder use does not eliminate the no-show problem

The reminder attendance rate is 47.63%, only slightly above the overall attendance rate of 46.28%. The reminder-sent group still contains a substantial proportion of no-shows.

Business implication: Reminders should not simply be treated as a one-size-fits-all solution. Their timing, channel, targeting, and frequency may need to be improved.

7. SMS shows the strongest attendance pattern among reminder channels

The reminder-channel visual shows SMS with the largest attendance proportion among the available reminder channels.

Business implication: SMS may be a useful channel for future reminder strategies, but further testing is required before concluding that it is more effective than other channels.

8. Gender shows relatively little difference in outcomes

Male and female patients show broadly similar attendance and no-show distributions.

Business implication: Gender alone does not appear to be a strong basis for prioritising no-show interventions.

BUSINESS RECOMMENDATIONS

Based on the findings, the following recommendations are proposed.

1. Develop Targeted Interventions for Repeat No-Show Patients

Patients with previous no-shows should be identified as a priority group for enhanced appointment engagement. Possible interventions include: additional confirmation messages, personalised reminders, earlier follow-up communication, and confirmation requirements for high-risk appointments.

2. Use Booking Lead Time to Guide Reminder Timing

Appointments booked 31–60 days in advance should receive appropriately timed reminders closer to the appointment date. This can help reduce the possibility of patients forgetting appointments or experiencing changes in plans during the waiting period.

3. Strengthen Reminder Strategies

The findings suggest that reminders alone do not eliminate no-shows. HealthConnect should evaluate: reminder timing, reminder frequency, reminder channel, message content, and patient risk characteristics.

SMS may be considered as one channel for further evaluation because it showed the strongest attendance pattern in the current analysis.

4. Pay Attention to Patients Travelling Long Distances

Patients travelling longer distances could receive additional support, such as earlier reminders, clearer appointment information, or flexible scheduling where operationally possible.

5. Monitor Excessive Waiting Times

HealthConnect should monitor appointments associated with longer waiting times and investigate the operational causes of delays. Reducing unnecessary waiting may improve patient experience and potentially support better attendance.

6. Develop a Future No-Show Risk Model

The variables identified in this analysis, including previous no-shows, booking lead time, distance, waiting time, appointment type, reminder status, and patient characteristics, could be considered as candidate features for future predictive modelling.

CROSS-TRACK COLLABORATION

Track Identified:

HealthConnect is a multidisciplinary project involving several technical tracks. During Week 5, the Data Analytics track identified a meaningful dependency with the Data Science track.

Information/Output Relevant to the Track

The Data Analytics track produced exploratory findings, KPIs, and identified relevant variables that may support future predictive analysis. These include appointment outcome, previous no-shows, booking lead days, reminder status, appointment type, age group, distance to clinic, and waiting time.

Why the Collaboration Is Relevant:

These analytical outputs can help the Data Science track understand the structure of the data base, identify potentially useful features, and determine areas that may require further investigation when developing a future model for predicting appointment no-shows.

Outcome

There was no direct conversation or information exchange with the Data Science track during Week 5. However, the Data Analytics work established a potential dependency subsequent project stages. The completed data preparation, exploratory analysis, and KPI development provide a foundation that can support future predictive modelling.

Further direct collaboration and exchange of requirements or outputs can be carried out in subsequent weeks as the multidisciplinary HealthConnect project progresses.

ASSUMPTIONS, LIMITATIONS, RISKS & DEPENDENCIES

The assumptions, limitations, risks, and dependencies identified during Week 4 were reviewed during Week 5 to determine whether they remained relevant, were resolved, or required modification.

Assumptions That Remain Relevant

The main assumptions from Week 4 remain relevant. The analysis assumes that the recorded appointment outcomes accurately represent whether appointments were attended, missed, or cancelled. It also assumes that the recorded patient, appointment, reminder, booking, and historical variables are sufficiently representative for exploratory analysis.

The analysis also assumes that relationships observed between variables and appointment outcomes represent associations rather than direct causal relationships.

Issues Resolved During Week 5

Several data-quality issues identified during the earlier analysis were addressed during Week 5. In particular, the appointment date and booking date fields were converted from text to date format to support appropriate analysis.

Duplicate and data-quality checks were completed, and no completely duplicated records or duplicate appointment IDs were identified.

Limitations That Remain

The dataset contains limited missing values in some variables, particularly distance to clinic and waiting time. These missing observations may slightly affect analyses involving those variables.

The dataset also does not capture every possible reason why a patient may miss or cancel an appointment. Factors such as unexpected emergencies, transportation problems, changes in personal circumstances, or other external factors may therefore not be reflected.

In addition, the dataset is fictional/anonymised, meaning that findings should be treated as analytical insights for the HealthConnect project rather than as evidence of actual patient behaviour

Risks

A major analytical risk is interpreting associations as evidence of causation. For example, the dashboard shows different appointment outcomes between reminder groups, but this does not prove that reminders caused the observed difference.

Another risk is that missing values or uneven group sizes may affect comparisons between certain categories.

Dependencies

The Data Analytics track depends on the availability and consistency of the HealthConnect dataset and data dictionary. Cross-track development between Data Analytics and Data Science will also depend on the analytical findings and relevant variables identified during this stage.

Future predictive modelling will depend on the quality and availability of the cleaned analytical dataset and on the variables identified during the exploratory analysis.

New Issues Identified During Week 5

No major new data-quality issues were identified during the Week 5 exploratory analysis beyond the limitations already documented. However, the analysis highlighted the need for further

investigation of the relationships between appointment outcomes and factors such as previous no-shows, reminders, booking lead time, appointment characteristics, waiting time, and distance to the clinic.

Potential Impact and Proposed Actions

These limitations and risks may affect the interpretation and generalisability of the findings. To address them, HealthConnect should:

- Continue monitoring data quality
- Investigate missing observations
- Avoid causal claims from descriptive analysis
- Validate important findings using additional data where available
- Examine group sizes when comparing categories
- Use future modelling or statistical analysis to test the strongest observed relationships

CONCLUSION

The Week 5 analysis provided a more detailed understanding of appointment attendance and no-show patterns within the HealthConnect dataset. The analysis found an overall 48.46% no-show rate, compared with a 46.28% attendance rate. Previous no-show history was particularly important, with a 55.41% repeat no-show rate among patients with previous no-shows.

The exploratory analysis also highlighted potential relationships involving booking lead time, distance to the clinic, waiting time, appointment characteristics, and reminder activity. A two-page Power BI dashboard was developed to communicate the KPIs and analytical findings.

Overall, the findings suggest that HealthConnect could benefit from targeted rather than uniform appointment-management strategies, particularly for patients with previous no-shows, longer booking lead times, longer travel distances, or other characteristics associated with poorer attendance.

The findings also provide a foundation for future predictive analysis and cross-track development within the wider HealthConnect project.